



SolarFlow PM

NEW FEATURES & AI AND AUTOMATION BLUEPRINT

E-signature, Ask SolarFlow & the road to zero-touch

What was built in this round, how it works and how to switch it on, and a prioritised plan of AI integrations and automations to take routine human work out of every stage.

Delivered

Attachments, PandaDoc e-signature, change orders, the AI assistant

How it works

Flows, rules, security, failure handling, setup

Principles

Guardrails that let automation act safely

48 recommendations

Stage by stage, with impact and effort

Zero-touch journey

Every human touch, today and after

Roadmap & cost

Phases, spend, risks, KPIs, your checklist

Table of contents

1	Summary	2	11	Survey and design	15
2	Stage attachments (replacing Drive Updated)	4	12	Permits and interconnection	16
3	E-signature for contracts	5	13	Procurement and install	17
4	Change orders	6	14	Inspection, PTO and after-sales	18
5	Setting up PandaDoc	7	15	Across the office: assistant, copilot, briefings and foresight	19
6	Ask SolarFlow — the assistant	8	16	The journey, touch by touch	20
7	Ask SolarFlow — security, cost and limits	9	17	Roadmap	22
8	Principles and the automation backbone	10	18	Cost, risk and governance	23
9	Documents and stage completion	12	19	Your checklist	25
10	Sales and signing	13			

How to read this document

Chapter 1 is the summary. Chapters 2–7 describe what was built in this round, how each part works, and how to switch it on. Chapters 8–15 are the blueprint: the rules that let automation act safely, then the recommendations stage by stage, each rated for impact **HIGH** **MEDIUM** and effort **S · M · L**. Chapters 16–19 put them together: the journey with its human touches before and after, the phased roadmap, cost and risk, and the checklist of things only you can do.

It sits alongside the main *Complete Project Documentation*, which has been updated for everything described here.

Summary

Four things were built in this round, each tested end to end. Behind them sits the plan this document exists for: a set of automations and AI integrations that take the routine work out of every stage, so that people handle exceptions and customers rather than paperwork.

Delivered

Stage attachments LIVE

The *Drive Updated* tick is gone. Each stage form now has an Attachments card, with that stage's files: survey photos, plan set, permit and ICA approvals, inspection report, PTO letter and so on. A project cannot advance until the required files are actually on it. Stages already closed under the old tick stay closed. A fix to stage-form uploads came with it.

PandaDoc e-signature LIVE · NEEDS KEYS

From the signing form, the contract goes to the homeowner by email, or is signed on screen with them present. When they sign, the project is created exactly as if the rep had pressed Sign, and the signed PDF is filed on it. Webhooks are verified, completion is idempotent, and every failure leaves a reason with a Finish button.

Change orders LIVE

A panel on every project: raise a change order with its reason and amount, send it for signature through PandaDoc, or approve it by hand. Once it is signed or approved, the amount is added to the contract value, once, and the signed PDF is linked from the change order.

Ask SolarFlow LIVE · NEEDS KEY

An AI assistant in the sidebar of every staff and dealer screen. Ask "give me the PM report" or "which projects are stuck in Permits, and why?" and it answers from live data. It runs on Claude Opus 5, reads through eight tools and never sees more than the asker could, because every lookup runs under that person's own database permissions.

Along the way, a real bug surfaced and was fixed: a sales rep's dealer list on the signing form was always empty, because the dealers table is closed to sales. A dealer directory now provides names and nothing else (migration 004500).

To switch them on

STEP	WHERE
Apply the database changes	Admin → Database → Apply: migrations 004300, 004400 and 004500.
PandaDoc API key and webhook key	Vercel → Environment Variables: PANDADOC_API_KEY , PANDADOC_WEBHOOK_KEY (both Sensitive). Chapter 5.
Contract and change-order templates	Built in PandaDoc; their IDs go in Admin → Settings → E-signature. Chapter 5.
Claude API key	Vercel: ANTHROPIC_API_KEY (Sensitive). Chapter 6.

The blueprint in one page

The recommendations follow one idea: **every stage should finish itself when the evidence arrives**. A document or a status update is read, checked against the stage's rules, and the project moves on, with a person asked only when the evidence is missing, unclear or contradicts itself. SolarFlow is well placed for this. The stage

rules, the permissions and the audit trail already live in the database, so an automation acting through them gets the same refusals a person would.

48

RECOMMENDATIONS

32 → 9

MANUAL STEPS PER PROJECT (CH.
16)

5

PHASES OVER ~9 MONTHS

The ten with the most value, in order

1. **Switch on the scheduler.** Reminders, the quiet-hours digest and report schedules are already written; they only need a cron (ch. 8).
2. **Document AI on every upload.** Classify each file, read its facts into the stage form (permit numbers, approval dates, PTO date, utility usage), and flag anything unreadable (ch. 9).
3. **Evidence-based auto-advance.** When the required fields and attachments are in and a document proves the milestone, advance the project without a click (ch. 9).
4. **Speed-to-lead in under 60 seconds.** Instant SMS and email with a booking link, round-robin assignment, and reminders with no-show recovery (ch. 10).
5. **Utility-bill reader to proposal.** A bill photo becomes usage, rate and utility on the deal, then a preliminary design and price (ch. 10).
6. **Auto-send the contract.** On *Financing approved*, send the PandaDoc contract built from the proposal, with reminders until signed (ch. 10).
7. **Permit and interconnection watcher.** Poll portals and inboxes for status changes, write them to the permit track, and chase what has gone quiet (ch. 12).
8. **Design-to-contract check.** Compare the final design with the signed system and raise the change order automatically when they differ (ch. 11).
9. **Ask SolarFlow, v2.** Let the assistant do things as well as answer (move a stage, ask the customer, raise a task), always with a confirm button (ch. 15).
10. **Homeowner copilot.** Draft or send replies to routine "where is my project?" questions from the project's own data, with the PM on the loop (ch. 15).

Stage attachments

"Drive Updated" was a promise that the files were somewhere else. Attachments are the files themselves, on the project, where the gate, the reports, the assistant and every future automation can see them.

What each stage asks for

STAGE	REQUIRED TO ADVANCE	OPTIONAL
Survey	Site survey photos	Survey report
Design	Plan set (PDF)	Stamped plans
Permits	Permit approval · ICA approval · HOA approval (not when the HOA track is N/A)	NTP approval
Procurement	Delivery confirmation	Purchase order
Install	—	Install sign-off (install pictures stay on the form's own field)
Inspection & PTO	Inspection report · PTO letter	Monitoring activation
Complete	—	Completion certificate / final documents

The rules

- **The gate names the file.** A refused advance says exactly what is missing ("Site survey photos not attached"), in the same words on the board, the project page and the stage form.
- **Old projects are not asked twice.** A stage closed under the old Drive Updated tick counts as complete. The tick is kept in the database but is no longer offered.
- **Hidden from the homeowner by default,** like every other document. The PM can share a file from the project's Customer panel.
- **Reports agree with the gate.** The report fields that used to read "Drive Updated (S1...S6)" now read "Attachments complete (S1...S6)", and "Final documents attached" for Complete. They are computed by the same rule the gate uses, old ticks included.

Why this matters for automation

With the files on the project, a document reader (chapter 9) can open the plan set or the PTO letter, pull out the facts, and close the stage without anyone opening a folder. A tick box never offered that.

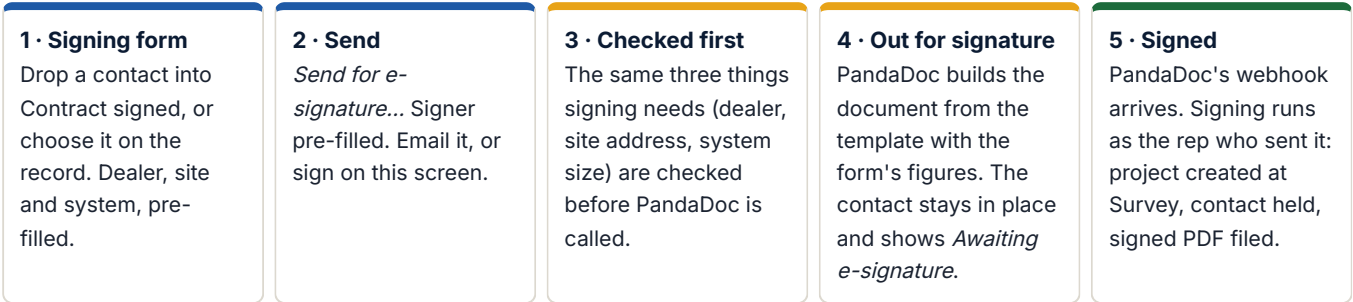
A fix that came with it

Testing showed that every stage-form upload had been failing: the upload function stored the file's kind as text in a column typed as a list of kinds. Migration **004300** fixes it, as a new file, because a database that already has the original would never see an edit to it.

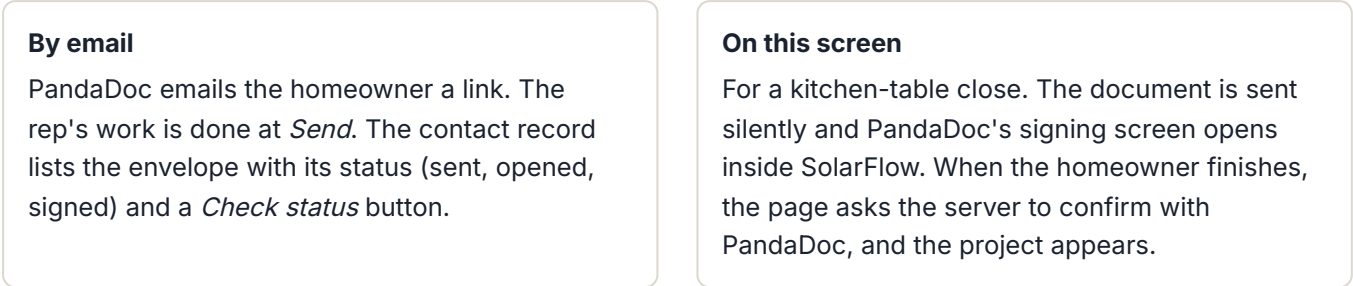
E-signature for contracts

The contract now goes to the homeowner from the signing form, and their signature is what creates the project.

The flow



Two ways to sign



What protects it

RISK	WHAT HAPPENS
A forged "signed" notice	The webhook is accepted only with PandaDoc's HMAC-SHA256 signature over the exact body, checked in constant time. A missing key refuses everything.
The same signature arriving three times	Webhook retries, on-screen signing and <i>Check status</i> may all report it. The first applies it; the rest change nothing.
No webhook configured	<i>Check status</i> asks PandaDoc directly and applies the result. The webhook is recommended, not required.
Signed, but it cannot apply	The signed PDF is kept regardless; the envelope shows the reason and <i>Finish</i> runs it again. A contact signed by hand meanwhile just gets the PDF on their project.
Two contracts out at once	Refused: one open envelope per contact. <i>Void</i> withdraws the old one in PandaDoc too.
The sender has since left	A webhook acts as the sender, with their permissions and name. If they are deactivated, the envelope waits for <i>Check status</i> .

Statuses: PREPARING SENT OPENED SIGNED DECLINED VOIDED FAILED (PandaDoc refused, in its own words). A signed envelope not yet applied shows in red with *Finish*.

Change orders

A change to a signed contract (a main panel upgrade, fewer panels, a critter guard) is now a record with a number, a reason, an amount and a signature, and the contract value follows it.

On the project page

1. **Raise:** + *New change order* with a reason, details and the change to the contract value (negative for a credit). Tick whether the homeowner must sign. It takes the company's next number (Admin → Settings: CO prefix and counter).
2. **Send for e-signature:** by email or on screen, as for contracts. Its status becomes *Out for signature*.
3. **Or approve by hand:** *Signed on paper* — *approve*, or *Approve* for one that needs no signature. Admin and ops only.
4. **Void** a draft or pending one. An approved change order cannot be voided; raise a new one to reverse it, so the history stays true.

When it is signed or approved

- The amount is added to the project's contract value, once, in the same database statement that approves it.
- The signed PDF is filed on the project and linked from the change order.
- The activity log records the change order, its number, its amount and whether it came through PandaDoc or by hand.
- A homeowner who declines leaves it *Declined*, and it can be revised and sent again.

Where this goes next

Today a person raises the change order. Chapter 11 recommends raising it automatically when the final design differs from the signed system, so the homeowner gets it minutes after the designer finishes.

Setting up PandaDoc

About half an hour, once. Until it is done the signing form simply does not offer e-signature, and everything else works as before.

Steps

#	STEP	HOW
1	Check the plan	API access and webhooks need a PandaDoc plan that includes the API. Confirm this with PandaDoc before building templates.
2	Apply the SQL	SolarFlow → Admin → Database → <i>Apply</i> (migrations 004400 and 004500).
3	API key	PandaDoc → Settings → Integrations → API → a production key. Vercel → Environment Variables → <code>PANDADOC_API_KEY</code> , Sensitive . Redeploy.
4	Contract template	Your installation agreement as a template with one signer role, <code>Client</code> (or your name for it, set in step 6), and tokens where the figures go, e.g. <code>[System.SizeKw]</code> . The list is below.
5	Change-order template	The same, with change-order tokens such as <code>[ChangeOrder.Amount]</code> .
6	Tell SolarFlow	Admin → Settings → E-signature: each template's ID (the end of its address in PandaDoc) and the signer role. The page shows whether the keys are set.
7	Webhook	PandaDoc → Integrations → Webhooks: the URL shown in Admin → Settings, event Document state changed . Its shared key goes in Vercel as <code>PANDADOC_WEBHOOK_KEY</code> (Sensitive). Redeploy.
8	Try it	Send a contract to yourself from a test contact and sign it; the project appears.

Fields the templates can use

Contract Client.FirstName · Client.LastName · Client.Email · Client.Phone · Project.Address · Dealer.Name · Company.Name · System.SizeKw · System.ModuleQuantity · System.BatteryQuantity · Contract.Value · Contract.GrossPrice · Contract.DownPayment · Contract.AmountFinanced	Change order Client.FirstName · Client.LastName · Client.Email · Project.Code · Project.Address · Company.Name · ChangeOrder.Number · ChangeOrder.Reason · ChangeOrder.Description · ChangeOrder.Amount · Contract.CurrentValue · Contract.NewValue
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Money arrives formatted (\$24,500.00). A token with no value is left out, so the template's own default text shows.

Who can do what

ACTION	ROLES
Send a contract; check, sign here, void	Admin, ops, sales: the people who can sign a contact by hand.
Raise, send, approve or void a change order	Admin and ops (the project team). Sales reps cannot see change-order envelopes at all.
Choose templates	Admin.

Ask SolarFlow

A question box on every staff and dealer screen. It answers from the live database, with the asking person's own permissions, in a few seconds. It is the first AI in the product that people use directly, and the base the later chapters build on.

What people ask it

ROLE	TYPICAL QUESTIONS
Admin · PM	"Give me the PM report." "Which projects are stuck in Permits, and why?" "What is blocking installs this week?" "Which customers are unhappy right now?" "What is still missing on PRJ-1A2B3C4D?"
Finance	"Pipeline value by stage." "Projects completed this quarter and the average days to complete." "Contract value by dealer for projects signed this year."
Sales	"Who is quoted but not contacted in 7 days?" "My deal pipeline by stage." "Which of my signed customers are in Survey?"
Dealer	"Where are all my projects?" "Which of mine are on hold, and why?" "What is still missing on my projects in Design?"

How it answers

The assistant is Claude Opus 5, given the question and a set of tools. It decides which lookups to run (often several at once), reads the results and writes the answer, with each project's code linked to its page. It runs up to eight rounds of lookups for a hard question. Progress shows while it works ("Building the PM report...").

TOOL	READS	OFFERED TO
find_projects	Projects with stage, days in stage, ageing past the threshold, what is missing to advance, PM, dealer, customer, unread customer messages, open follow-ups. Filters by stage, status, PM, dealer, customer, jurisdiction, ageing or blocked.	Everyone
project_details	One project in full: people, system, stage history with dates, missing items, change orders, documents, open requests to the customer, ratings, recent activity.	Everyone
pm_report	Per PM: active and on-hold counts, ageing, blocked and why, messages waiting, follow-ups, and each project.	Admin, ops
dashboard_summary	The dashboard for a period: headline numbers, projects per stage, the needs-attention lists, PM workload.	Admin, ops, finance
run_report	The report builder itself: columns, group-by, sums and averages, filters, with that role's field list and money rules.	Admin, ops, finance
find_contacts · find_deals	Contacts by stage, owner and days since contact; deals with value, probability and next action, plus value per stage.	Admin, ops, sales
customer_feedback	Ratings per stage and party, response rates, recent comments.	Admin, ops

Security, cost and limits

Why it cannot leak

Same permissions as the screens

Every tool reads through the asker's own database session, so row-level security decides what comes back. A dealer asking about "all projects" gets their own book. Asking by another dealer's project code returns "no project you can see".

Read-only

No tool writes. Asked to move or change something, it says where in the app to do it. Chapter 15 describes how actions could be added safely.

Offered only what the role has

Sales is not offered the dashboard, and dealers get two tools. Money fields follow the report builder's rules, so a contract value reaches only admin, ops and finance.

Safe on the page

Answers render as formatted text built element by element; no HTML from the model reaches the page. A link is live only if it points inside SolarFlow.

- **Logged:** every question is written to the activity log (`assistant.asked`) with who asked, which tools ran, the time taken and the tokens used. The answers are not stored.
- **Limited:** 40 questions per person per hour. Tool results are capped in size, so a runaway query cannot flood the model.
- **Refusals:** server-side fallbacks are on. If the model's safety filter declines a request, the API retries it on the model Anthropic recommends for that case, and a final refusal becomes a polite sentence.
- **Conversation:** kept in the browser tab only, and sent with each question. Closing the tab forgets it.

Cost

Claude Opus 5 costs \$5 per million input tokens and \$25 per million output tokens; the fixed instructions are cached at a tenth of that. A typical question runs one or two lookups.

USAGE (ESTIMATE)	APPROXIMATE COST
One simple question	About 2–5 US cents
A PM report or a multi-step question	About 5–15 US cents
20 people × 10 questions a working day	About \$10–30 a day, \$200–600 a month

These are estimates. The activity log records the real tokens for every question, so after a week the actual figure can be read off (a report on `assistant.asked`). Two settings trade quality for cost: `ASSISTANT_EFFORT` (default `medium` ; `low` is faster and cheaper, `high` is more thorough) and `ASSISTANT_MODEL` . Test a cheaper model on your own real questions before switching.

Limits today

- It reads, and does not act.
- A project list covers the 300 most recent; the report builder has no such limit.
- It knows what the database knows: a permit status nobody entered cannot be found (chapters 9 and 12 fix that at the source). A long question can take up to 120 seconds on Vercel.

Principles and the automation backbone

Cutting human work does not mean trusting a model with everything. It means building a system that does the routine work itself, knows when it is unsure, and puts those cases in front of a person with everything needed to decide in one click.

Seven rules, and the idea behind them

1 • Plain automation first, AI second

If a rule can decide it (a date passed, a file arrived, a status changed), a rule decides it. AI is for reading documents, understanding messages and drafting, where rules cannot reach. Rules are cheaper, faster and never surprising.

2 • The database is the guardrail

Automations act through the same functions and permissions as a person. The stage gate, the hold, the won-deal lock and row-level security refuse an automation exactly as they would refuse a PM. Nothing gets a back door.

3 • Confidence decides who acts

Every AI result carries a confidence score. Above the line, the system acts. Below it, the result goes to the *exceptions* queue (the table already exists) as a suggestion a person accepts, edits or rejects. The line is set per task, and moves as accuracy is proven.

4 • Evidence, not assertion

A stage advances because a document proves the milestone (the PTO letter with its date) and the fact is recorded with a link to that document. Anyone can see why the system did what it did.

5 • Every action is logged and reversible

Automations write to the same activity log as people, under a named system actor, with the rule or model that acted and its confidence. A wrong move is undone like any other.

6 • Idempotent and retry-safe

Webhooks repeat, jobs rerun and networks fail. Every job can run twice without doing anything twice. The e-signature completion is the pattern to copy.

7 • Measured before it is trusted

Each AI task has a test set of real, labelled examples (fifty bills, fifty permits). A task runs on its own only once it passes on that set, and the set is re-run whenever the model or the prompt changes.

Human on the loop, not in it

People stop being the transport between systems (reading an email and typing it into a form) and become the reviewers of the few cases the system flags. The PM's day starts from the exceptions list, not from the inbox.

Autonomy levels

LEVEL	NAME	WHAT THE SYSTEM DOES
L0	Manual	People do it; the system records it. (Much of the pipeline today.)
L1	Assisted	The system reminds, checks and answers questions. (Stage gates, reminders, Ask SolarFlow.)

LEVEL	NAME	WHAT THE SYSTEM DOES
L2	Drafts	The system prepares the work: fields filled from a document, a reply drafted. A person approves it.
L3	Acts, flags doubt	The system acts on high-confidence cases and routes the rest to exceptions.
L4	Zero-touch	The system handles the whole step. People see a summary and the rare exception.

The backbone to build first

Five shared pieces that every later recommendation reuses. They are not features anyone will see, but without them each automation would be built from scratch.

PIECE	WHAT IT IS	WHY
B1 Scheduler & job queue HIGH S	Vercel Cron calling the existing <code>/api/push/reminders</code> every few minutes, plus a small job table (<code>jobs</code> : kind, payload, run_at, attempts, last_error) processed by the same cron. Jobs are idempotent and retried with backoff.	Install reminders, the quiet-hours flush, the PM digest and report schedules are already written and simply never run.
B2 Event outbox & webhook inbox HIGH M	Database triggers write domain events (<code>project.stage_changed</code> , <code>document.uploaded</code> , <code>deal.stage_changed</code>) to an outbox table; the job queue fans them out to automations. Inbound webhooks (PandaDoc, lenders, monitoring, SMS replies) land in one verified inbox route.	Automations react to what happened instead of polling, and one event can drive several of them without coupling them together.
B3 Exceptions queue screen HIGH S	A screen over the existing <code>exceptions</code> table: what the system proposed, the evidence (the document, highlighted), its confidence, and Accept / Edit / Reject. Grouped by PM, with counts on the dashboard.	This is what lets automation act alone on the clear cases: the unclear ones have somewhere to go.
B4 AI gateway HIGH S	One server module (grown from <code>lib/assistant</code>) through which every model call passes: model choice per task, prompt caching, structured outputs validated against a schema, cost and latency logged per task, retries and fallbacks, and the per-task test sets.	One place to see spend, swap models and prove quality. No model call is scattered through the code.
B5 Communications hub HIGH M	SMS (e.g. Twilio) beside the existing email and push, with templates, consent and quiet hours in one place, and two-way SMS threaded into the project chat and the contact's activity log.	Most homeowner and lead contact happens by text. Speed-to-lead, reminders and the homeowner copilot all need it.

Documents and stage completion

Most of a PM's time goes on moving information from documents into forms. Now that every stage has its attachments (chapter 2), the documents are in one place for a machine to read.

RECOMMENDATION	HOW IT WORKS IN SOLARFLOW	PEOPLE AFTER
D1 Classify every upload HIGH S	On <code>document.uploaded</code> , a vision model reads the first pages and labels the file: permit approval, ICA, PTO letter, plan set, utility bill, inspection report, HOA letter, other. A mislabelled or generic upload (a homeowner's "photo") is refiled under the right stage category. A cheap model is enough here.	Nobody sorts files. Low-confidence labels go to exceptions.
D2 Read the facts into the form HIGH M	Per document type, a structured-output prompt extracts the fields the stage form already has as typed columns: permit number, issue and expiry dates, AHJ; ICA approval date; PTO date; inspection result and date; plan-set system size and module count. Values are written with <code>source = 'ai'</code> , the document linked, and a confidence score.	The PM reviews only fields below the confidence line, beside the highlighted document.
D3 Evidence-based auto-advance HIGH S	After D2 writes a fact, the job re-evaluates the stage (<code>evaluateStage</code> , the same gate people use). If nothing is missing and the evidence is high-confidence, the project advances, with a note naming the document that proved it. Configurable per stage, starting with Inspection & PTO and Permits.	The PM stops clicking "advance". They see a feed of what moved and why.
D4 Photo quality check HIGH M	Survey and install photos are checked against a shot list (roof planes, main panel with the label legible, meter, attic, conduit run, inverter and labels, array complete). A missing or blurry shot is flagged while the crew is still on site, via SMS to the surveyor or installer.	No return visits for a missing photo. The PM sees only failures.
D5 Missing-document chaser MEDIUM S	When a stage waits on a document only the homeowner, dealer or utility has (an HOA approval, a bill, the dealer code form), the system uses the existing <i>Ask the customer</i> feature, or emails the dealer, on a schedule: day 0, day 3, day 7, then a task for the PM.	The PM steps in only after the third reminder.
D6 Expiry and validity watch MEDIUM S	Dates read by D2 (permit expiry, a lender approval's expiry, the NTP validity) are watched by B1. A warning goes to the PM 14 days before and a task 3 days before, with the project flagged on the board.	No expired permits found on install day.

Why this comes first

D1–D3 turn every later integration into a source of documents. A permit portal that cannot be connected can still email a PDF, and once the PDF is read the stage completes itself. It is the widest-reaching change for the effort.

Sales and signing

In residential solar the first minutes after an enquiry decide a large share of the outcome, and most of the sales team's administration is copying: from a web form into a contact, from a bill into a deal, from a proposal into a contract.

RECOMMENDATION	HOW IT WORKS IN SOLARFLOW	PEOPLE AFTER
S1 Web and ad leads straight in HIGH S	Website forms and ad platforms (Meta and Google lead forms) post to a signed endpoint that calls the existing <code>create_contact</code> , with UTM source, medium and campaign. The duplicate check across every email and phone runs as it does today.	No re-typing. Suspected duplicates go to exceptions.
S2 Speed-to-lead HIGH S	On <code>contact.created</code> , within 60 seconds, an SMS and email go out (B5) with the rep's name and a booking link. If nothing is booked in 15 minutes, a call task goes to the owner. All inside the contact's consent and quiet hours.	The rep calls only the leads who have not booked themselves.
S3 Routing MEDIUM S	The owner is chosen by rules: ZIP-code territory, language, dealer, rep capacity (open deals) and availability, round-robin within a team. The owner field already exists; today it defaults to the creator.	No morning lead-assignment meeting.
S4 Self-booking, reminders, no-show recovery HIGH M	A booking page over the reps' calendars (e.g. Google Calendar or Cal.com). Booking moves the contact to <i>Appointment scheduled</i> . Reminders go 24 hours and 2 hours before, each with a reschedule link, which moves the contact to <i>Rescheduled</i> . A missed appointment moves it to <i>No-show</i> and starts a rebooking sequence. The contact stages board already has these columns.	No phone tag. The board moves by itself.
S5 Utility-bill reader HIGH S	The bill photos already uploaded on the deal (<code>electricity_bill_front/back</code>) are read by a vision model into the deal's typed fields: utility, rate plan, monthly and annual kWh, average bill. Structured output is validated against the schema.	The rep confirms low-confidence reads only.
S6 Instant preliminary design and price HIGH L	Address plus usage go to a roof and sunlight service (e.g. Google Solar API, or a design tool's API such as Aurora or OpenSolar), which returns the usable roof, module count, system size and estimated production. The price book and adder rules (both tables exist) produce the price, and the result becomes a draft proposal version on the deal.	The rep reviews and presents it; the designer finalises after signing.
S7 Financing status sync HIGH M	The lender application link is sent from the deal. The lender's webhook or API (e.g. GoodLeap, Mosaic, Sunlight, LightReach) writes the decision, amount and terms to the deal, and an approval moves the contact to <i>Financing approved</i> . Stipulations become asks to the homeowner.	Nobody checks lender portals.

RECOMMENDATION	HOW IT WORKS IN SOLARFLOW	PEOPLE AFTER
S8 Contract out automatically HIGH S	Built on the new e-signature: when a deal reaches <i>Financing approved</i> (or cash with a deposit) and its proposal is accepted, the PandaDoc contract is sent with the proposal's system. Reminders go out on days 1, 3 and 5 if it is unsigned, then a task for the rep. Signing creates the project as it does today.	The rep no longer prepares or chases contracts.
S9 Call and meeting notes MEDIUM M	Recorded calls, or the rep's voice memo after an appointment, are transcribed and summarised into the activity log: objections, next step, decision makers, roof or shading notes, competitor. Any agreed follow-up date becomes the deal's next action.	Reps stop writing notes; the next action is never blank.

Survey and design

RECOMMENDATION	HOW IT WORKS IN SOLARFLOW	PEOPLE AFTER
SD1 Survey books itself HIGH M	On project creation at Survey, the homeowner gets a booking link over the surveyors' availability slots (the <code>availability_slots</code> table exists), grouped by area to limit driving. Booking fills the survey date. Reminders and rebooking work as in S4.	No scheduling calls.
SD2 Remote pre-survey MEDIUM M	Before anyone drives out, aerial imagery (e.g. EagleView, Nearmap or Google Solar API) gives the roof planes, pitch, obstructions and shading. Simple roofs with a clear main-panel photo from the homeowner go straight to design. Complex roofs get a site visit.	Surveyors visit only the roofs that need it.
SD3 Adders from the facts HIGH S	Survey facts (main panel rating read from the photo, roof type and age, trenching distance) run through the adder rules (<code>adder_rules</code> exists): a 100 A panel adds the upgrade. Adders are written with <code>source = 'rule'</code> or <code>'ai'</code> , and anything above a price limit needs approval.	Nobody hunts for adders at the end.
SD4 Design through the design tool's API MEDIUM L	The survey data opens a project in the design tool (e.g. Aurora or OpenSolar) through its API. The finished design (layout, equipment, production, plan set) comes back into the <code>designs</code> tables and the plan-set attachment.	Designers design; nobody copies figures across.
SD5 Plan-set review HIGH M	Before the plan set goes to the engineer or the permit office, a model checks it against a per-jurisdiction checklist (setbacks, labels, rapid shutdown, conductor sizing notes, required pages) and against the survey (module count, roof planes, main panel). Issues go back to the designer as a list.	Fewer permit rejections; the designer fixes a list rather than a rejection letter.
SD6 Design-to-contract check and automatic change order HIGH S	When the final design is approved, its system size, module count, battery and adders are compared with what was signed. A price-changing difference raises a change order with the reason and amount filled in and sends it through PandaDoc (chapter 4), and procurement waits for it.	No install built to a system the customer did not sign for.

Permits and interconnection

The longest wait in most projects, and the one people check most often. The work is rarely hard. It is repetitive: filling the same forms, then checking portals until something changes.

RECOMMENDATION	HOW IT WORKS IN SOLARFLOW	PEOPLE AFTER
P1 Jurisdiction knowledge base HIGH M	Per jurisdiction and utility (both tables exist): the required documents, fees, submission method, typical turnaround (measured from our own permit history) and common rejection reasons. It is kept current by the people who hit a change, and read by P2, SD5 and Ask SolarFlow.	The knowledge stops living in one person's head.
P2 Permit packet assembly HIGH M	From P1's list, the system gathers the plan set, stamped plans, spec sheets and site plan, fills the jurisdiction's application form from project data, and produces the packet ready to submit, or submits it where the portal has an API.	The permit coordinator checks and submits in minutes.
P3 Instant permits where available HIGH M	For jurisdictions on SolarAPP+ (automated permitting for standard residential systems), qualifying projects go through it and get the permit the same day.	Zero touches for qualifying systems.
P4 Status watcher HIGH M	Each open permit, ICA and HOA track is watched: portal APIs where they exist, a monitored inbox (the permit office's and utility's emails, read by D1 and D2), and a scheduled check on quiet portals. Changes write <code>permit_events</code> and the stage-form status. A track silent for longer than P1's typical turnaround gets a follow-up email and a task.	Nobody checks portals. The PM sees changes and overdue items.
P5 HOA packet and letter MEDIUM S	When the HOA track is required, a letter and packet are generated (site plan, array render, spec sheets) and sent to the homeowner to forward, or to the HOA directly. The approval comes back through D5 and D2.	No hand-built HOA packets.
P6 Rejection triage MEDIUM S	A rejection or correction letter is read and turned into a list of changes, assigned to design or to the permit coordinator, with the revision notes filled in on the permit track. Each rejection reason is added to P1 so the next packet avoids it.	Corrections start the same day the letter arrives.

Procurement and install

RECOMMENDATION	HOW IT WORKS IN SOLARFLOW	PEOPLE AFTER
PI1 Bill of materials and purchase order HIGH M	The approved design produces the bill of materials (<code>bom_items</code> exists): modules, inverters or optimisers, racking by roof type, wire and conduit by run length, and consumables. Quote requests go to distributors by email or API (<code>vendor_quotes</code> exists), and the best quote becomes a draft purchase order.	The procurement manager approves the PO.
PI2 Delivery tracking MEDIUM S	Order confirmations and shipping notices from distributors are read (D1 and D2) into the procurement stage: requested and delivered dates, with the delivery confirmation filed as the stage's attachment. That completes the stage.	Procurement closes itself when the goods arrive.
PI3 Crew scheduling HIGH L	Projects ready to install (permit in hand, materials delivered or dated) are proposed into crew calendars by crew skills (battery, tile, ground mount), drive time and the weather forecast. The homeowner confirms the date through a link, and the existing 48-hour and 24-hour reminders follow.	The install manager adjusts a proposed schedule instead of building one.
PI4 Day-before readiness check HIGH S	At 2 pm the day before each install: permit approved and not expiring, materials on site or on the truck, homeowner confirmed, weather acceptable, no open change order unsigned. Anything failing alerts the install manager while there is still time.	No truck rolls to a job that cannot start.
PI5 Install completion packet MEDIUM S	The crew uploads photos from the job (D4 checks them against the install shot list) and the homeowner signs off on a phone. Completion fills the install stage, files the sign-off and requests the inspection (I1).	The PM does not chase the crew for photos.

Inspection, PTO and after-sales

RECOMMENDATION	HOW IT WORKS IN SOLARFLOW	PEOPLE AFTER
I1 Inspection booking and result HIGH M	Install completion requests the inspection through the jurisdiction's portal or by email, using P1's method. The result (a card photo, an email or a portal status) is read into the stage. A failure opens a task with the inspector's notes, split into items.	Only failed inspections need a person.
I2 PTO letter completes the project HIGH S	The utility's PTO letter or email is read (D2) for the PTO date. With the inspection passed and monitoring active, the project advances to Complete (D3), and the homeowner gets the "power on" message that already exists.	Zero touches for a clean PTO.
I3 Monitoring activation check HIGH M	The inverter or monitoring platform's API (e.g. Enphase Enlighten, SolarEdge, Tesla) confirms the site is reporting and producing, and fills the monitoring and energisation fields. A site that has stopped reporting is flagged.	No "is it on?" calls.
I4 Production watch and service tickets MEDIUM M	After completion, daily production is compared with the design's estimate adjusted for weather. A sustained shortfall or an offline system opens a service ticket with the likely cause, and the homeowner gets a message before they notice themselves.	Service reaches the customer first.
I5 Reviews and referrals at the right moment MEDIUM S	A homeowner whose stage ratings are high gets a review request after "power on", and a referral invitation after the first strong production month. A referral arrives as a contact with the referrer as its source. Unhappy customers are excluded; the follow-up flow already catches them.	Marketing runs itself from delivery.

Across the office

The stage automations take work away. These make the remaining work faster: one place to ask and act, replies drafted from the project's own facts, and warnings before a project goes wrong rather than after.

RECOMMENDATION	HOW IT WORKS IN SOLARFLOW	PEOPLE AFTER
O1 Ask SolarFlow can act, with a confirm HIGH M	Add action tools to the assistant: move a stage, put on hold with a reason, ask the customer for something, raise a task or change order, assign a PM, send the contract. Each action returns a proposal card ("Move PRJ-1A2B to Design? The survey photos are in.") and runs only when the person presses Confirm, through the same functions and permissions as the screens, logged as done by that person through the assistant.	"Put the Smith job on hold until the HOA answers" is one sentence and one click.
O2 Homeowner copilot HIGH M	When a homeowner writes in the project chat, a reply is drafted from the project's stage, dates, missing items and typical durations, in the company's tone. Start at L2 (the PM edits and sends). Move routine questions ("when is my install?", "what is PTO?") to L3 once the PM accepts most drafts unchanged. Complaints, money and anything unusual always go to the PM.	The PM answers the few messages that need a person; the rest are answered in minutes, even at night.
O3 Morning briefing per PM HIGH S	At 7 am each PM gets a short briefing (email, push, and the top of their dashboard): what moved overnight and why, what is ageing past threshold, what is blocked and on whom, customer messages waiting, today's installs and inspections with their readiness, and the exceptions to clear. It is built from the same tools as Ask SolarFlow.	The day starts with a list, not an inbox.
O4 Delay and cancellation early warning MEDIUM M	From the project's own history (days in stage compared with typical, a stage revisited, a low rating, unanswered messages, a financing stipulation, a quiet homeowner), each active project gets a risk score and its top reasons. The highest risks are flagged on the board, and a retention call is suggested before the customer cancels.	PMs spend time on the projects at risk, not the loudest ones.
O5 Dealer and rep insights MEDIUM S	A monthly written summary per dealer and per rep, from the report builder: volume, conversion, cancellations and their reasons, cycle times, customer ratings, and changes from last month. Written in plain language, with the numbers linked to the report that produced them.	Partner reviews are prepared automatically.
O6 Cash and capacity forecast MEDIUM M	Milestone payments (down payment, M1-M3, finance M1/M2) projected from each project's stage and the typical time to reach each milestone, giving expected cash by week. Installs forecast the same way give crew capacity needed by week.	Finance and scheduling plan from a forecast instead of a spreadsheet.

The journey, touch by touch

Every routine thing a person does to move one project from enquiry to power-on today, and who does it once the blueprint is in place. Steps that become automatic still report what they did. "Exceptions" means a person sees it only when the system is unsure.

#	STEP	TODAY	AFTER	BY
1	Enter a web or ad lead	Person	AUTOMATIC	S1
2	Assign the owner	Person	AUTOMATIC	S3
3	First response	Person	AUTOMATIC	S2
4	Book the appointment	Person	AUTOMATIC	S4
5	Remind, reschedule, recover no-shows	Person	AUTOMATIC	S4
6	Hold the appointment and sell	Person	PERSON	Rep (notes by S9)
7	Type the bill's usage into the deal	Person	EXCEPTIONS	S5
8	Build the proposal	Person	PERSON REVIEWS	S6 drafts
9	Chase the lender's decision	Person	AUTOMATIC	S7
10	Prepare and send the contract	Person	AUTOMATIC	S8 (on the new e-signature)
11	Chase the signature	Person	AUTOMATIC	S8
12	Record the signing, create the project	Person	AUTOMATIC	E-signature (built)
13	Schedule the survey	Person	AUTOMATIC	SD1
14	Visit the site	Person	PERSON, FEWER	Surveyor (SD2 skips simple roofs)
15	Check the survey photos are complete	Person	EXCEPTIONS	D4
16	Work out the adders	Person	EXCEPTIONS	SD3
17	Design the system	Person	PERSON	Designer
18	Copy design figures into the project	Person	AUTOMATIC	SD4
19	Review the plan set	Person	EXCEPTIONS	SD5
20	Raise the change order	Person	AUTOMATIC	SD6 (on the new change orders)
21	Assemble and submit the permit packet	Person	PERSON CHECKS	P2 assembles; P3 where available
22	Check permit and interconnection portals	Person	AUTOMATIC	P4
23	Type approval details into the form	Person	EXCEPTIONS	D2
24	Chase the HOA	Person	AUTOMATIC	P5, D5
25	Build the bill of materials and order	Person	PERSON APPROVES	P11 drafts

#	STEP	TODAY	AFTER	BY
26	Track the delivery	Person	AUTOMATIC	PI2
27	Schedule the crew	Person	PERSON ADJUSTS	PI3 proposes
28	Readiness calls the day before	Person	AUTOMATIC	PI4
29	Install	Person	PERSON	Crew
30	Chase install photos and sign-off	Person	AUTOMATIC	PI5, D4
31	Book the inspection, record the result	Person	EXCEPTIONS	I1
32	Record PTO, check monitoring, complete	Person	AUTOMATIC	I2, I3, D3
+	Clear the exceptions queue (new, daily)	—	PERSON	B3

The count

Today all 32 steps take a person. Afterwards, eight of them still do (selling, the site visit where it is needed, design, checking the permit packet, approving the order, adjusting the schedule, the install itself and reviewing the proposal), plus one new daily step: clearing the exceptions. That is **32 manual steps down to 9**. Every one of the nine is judgement or physical work, not copying. Answering homeowners' routine questions (O2) comes off the list on top of this.

Roadmap

Ordered by value for effort, with the shared pieces first so that each later item is small. The dates are indicative, for one engineer working with the tooling this project already uses. Each phase ends with its items live and measured.

PHASE	WHEN	ITEMS	YOU WILL SEE
0 · Switch on	Weeks 0–2	PandaDoc and Claude keys and templates (chapters 5–6). B1 scheduler. Neon hygiene: default branch, rotated password, Sensitive variables. CI for the test suites.	E-signature and the assistant in use; reminders and digests running.
1 · Backbone and first wins	Months 1–2	B2 events, B3 exceptions screen, B4 AI gateway, B5 SMS · D1–D3 on Permits and Inspection & PTO · S1, S2, S3, S8 · P4 (monitored inbox) · I2	Leads answered in a minute; contracts go out by themselves; PTO letters complete projects.
2 · Documents everywhere	Months 2–4	S4, S5, S7 · D4, D5, D6 · SD1, SD3, SD6 · P1, P6 · PI4 · O1 (assistant actions) · O3 (morning briefing)	Self-booking; bills read; surveys self-scheduled; change orders raised automatically.
3 · Drafts become actions	Months 4–7	S6 · SD2, SD5 · P2, P5 · PI1, PI2, PI5 · I1, I3 · O2 (copilot, drafting, then answering routine questions)	Permit packets and orders prepared for approval; inspections booked; routine homeowner questions answered.
4 · Near zero-touch	Months 7–9+	SD4 · P3 · PI3 · I4, I5 · S9 · O4, O5, O6	Designs flow in; crews scheduled; service and referrals automatic; forecasts.

Dependencies to arrange early

- **Accounts and API access:** an SMS provider; calendar access for reps and surveyors; lender partner APIs or webhooks; the design tool's API; imagery provider; inverter monitoring API; distributor ordering (email is enough to start).
- **Test sets:** collect 50 real examples of each document type (bills, permits, ICAs, PTO letters, inspection cards, plan sets), with the right answers, before each document task is built.
- **A named owner** for the jurisdiction knowledge base (P1) and for the exceptions queue in each team.

Cost, risk and governance

Model cost (estimates)

List prices per million tokens: Claude Opus 5 \$5 in / \$25 out; Claude Sonnet 5 \$2 / \$10; Claude Haiku 4.5 \$1 / \$5. Cached prompt reads are about a tenth of the input price. Start every task on Opus 5, the most capable model. Move a high-volume task to a cheaper model only once its test set shows the cheaper model is just as accurate.

TASK	PER ITEM (ESTIMATE)	NOTES
Classify an upload (D1)	under 1 cent	A good candidate for a cheaper model once measured.
Read a document into fields (D2, S5)	about 2–4 cents	A page or two of image plus a short structured answer.
Review a plan set (SD5)	about 25–60 cents	A 20–40 page PDF; once per design revision.
Draft a homeowner reply (O2)	about 2–5 cents	Project facts plus the thread.
Morning briefing (O3)	about 10–20 cents per PM per day	Several lookups, one written summary.
Ask SolarFlow question	about 2–15 cents	Chapter 7.
A whole project	roughly \$2–5	About 30 documents, a plan set, 20 messages and a share of briefings. Small against the office hours it replaces.

All AI calls go through the gateway (B4), which records tokens and cost per task, so these estimates are replaced by real figures within weeks. Third-party services (SMS, imagery, design tools, permitting) are priced by their vendors and are not included.

Risks and controls

RISK	CONTROL
A wrong value read from a document	Schema-validated output; confidence threshold per field; the source document linked to every AI-written value; low confidence goes to exceptions; the stage gate still applies; accuracy tracked from accept/edit/reject.
Instructions hidden in a document or message	Documents and messages are treated as data, never as instructions. Document-reading tasks have no tools that act; they return a schema, not commands. Actions only ever come from rules or from a person's Confirm.
Too much automation too soon	A per-stage on/off switch and threshold for every automation; start at L2; a weekly sample audit of automatic actions; one-click undo from the activity log.
Privacy	Send the model only the fields a task needs; keep keys server-side; confirm the model provider's data-use and retention terms suit the business; no customer data in logs beyond what the activity log already holds.
Messaging consent	SMS only with recorded consent; quiet hours; STOP honoured at once; the consent basis kept on the contact, as for e-book subscribers.

RISK	CONTROL
Model changes	Model IDs pinned per task; the test set re-run before any change; the gateway can switch a task's model without code changes.
Cost overrun	A per-task budget and alerts in the gateway; rate limits per person (as the assistant already has); monthly cost report by task.
Vendor dependence	Each outside service sits behind one module (as PandaDoc does in <code>lib/esign</code>), so it can be replaced without touching the rest.

How to know it is working

MEASURE	TARGET DIRECTION
Manual steps per project	32 → 9. Measured from the activity log (person or system actor).
Stages advanced automatically	Rising each phase; Inspection & PTO first.
Exceptions per project · time to clear	Falling as thresholds are tuned; cleared the same day.
AI accuracy per task	At least as good as a person on the test set before running alone; accept-unchanged rate above 95%.
Speed to lead	Median under 1 minute.
Signed to install · install to PTO	Days falling; permit first-time approval rising.
Customer rating per stage · cancellations	Rising · falling.
Model cost per project	Within the estimate above, reported monthly.

Your checklist

The things only you can do, in order. The first block switches on what is already built.

Now

1. Admin → Database → **Apply** (migrations 004300, 004400, 004500).
2. PandaDoc: confirm the plan includes the API; create the API key; build the contract and change-order templates; add the webhook. Put the keys in Vercel as Sensitive and the template IDs in Admin → Settings (chapter 5).
3. Anthropic: create an API key at console.anthropic.com and add `ANTHROPIC_API_KEY` to Vercel as Sensitive (chapter 6).
4. Redeploy, then open `/api/health`. It should list both PandaDoc keys and the Anthropic key as present, and `migrations.behind` as empty.
5. Send yourself a test contract and a test change order; ask Ask SolarFlow for the PM report.

Carried over from the last review

1. Rotate the database password in Neon; the old one was shared in a chat.
2. Disconnect the Neon integration in Vercel and set `DATABASE_URL` by hand as Sensitive; make the branch holding the live data Neon's default and protect it.
3. Add the scheduler (B1): a Vercel Cron entry calling `/api/push/reminders` with `CRON_SECRET`.
4. Set the SMTP and VAPID variables so email and push reach people.

To start the blueprint

1. Choose an SMS provider and sign up (B5, S2).
2. Collect 50 real examples of each document type with the right answers (the test sets for D1–D2).
3. List the lenders, design tool, imagery provider, monitoring platforms and distributors you use, so the integrations can be ordered by volume.
4. Name an owner for the exceptions queue in each team, and for the jurisdiction knowledge base.